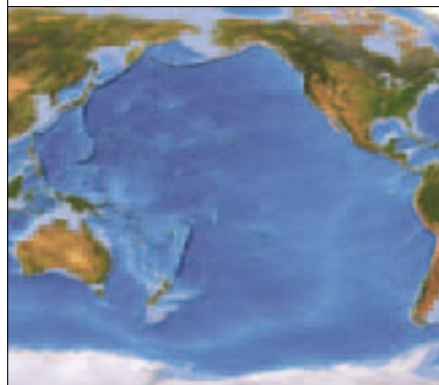




◀ EARTHQUAKE ZONES

The subduction zones are notorious for their earthquakes, caused by the sudden fracture of the rocks along the mobile plate boundary. The islands of Japan lie in one of these regions, along a boundary marked by the deep Japan Trench. Here, earthquakes are a daily event, with about a thousand tremors each year. Every few years a really big earthquake causes destruction on a massive scale, as in the Kobe earthquake in 1995.

PACIFIC RING OF FIRE



The Pacific is surrounded by ocean trenches, shown on this satellite image as dark lines extending from New Zealand to South America, and hundreds of volcanoes known as the Pacific Ring of Fire. Steady destruction of the ocean floor in such zones is shrinking the Pacific by 1 sq mile (2.5 sq km) a year.



Volcanic rock is fragmented into ash by the force of the explosion

▲ VOLCANIC CATAclysms

The lava produced by volcanoes in the subduction zones is much stickier than the lava erupted at midocean ridges. It can block the vent, building up pressure and causing explosive eruptions like that of Mount Pinatubo in the Philippines in 1991.